

Cartesian Bicategories

and Cartesian Equipments

Categorical Logic Meets Double Categories – Side B



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This talk is based on my recent thesis :

Logical Aspects of
Virtual Double Categories

Chapter 2 :

Categorical Logic meets
Double Categories

Slides



Outline

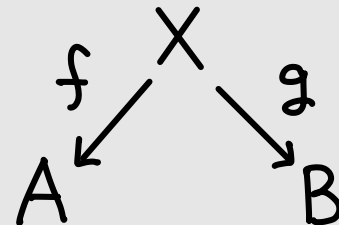
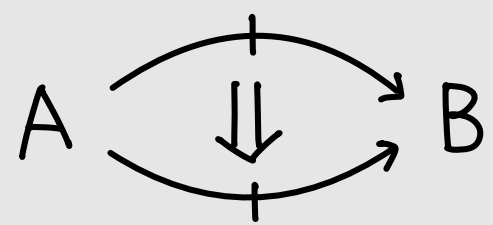
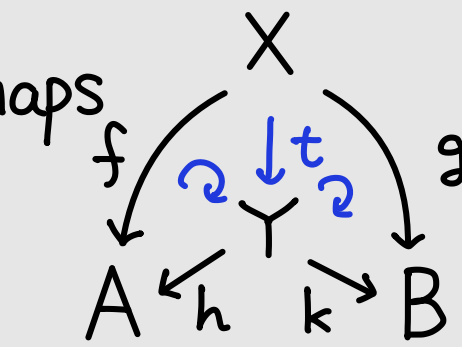
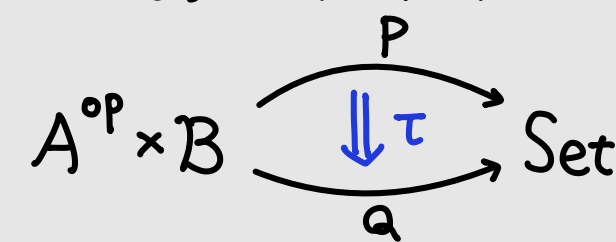
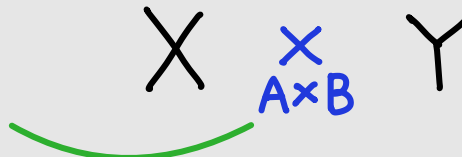
1. Cartesian bicategories
 2. Cartesian equipments
 3. Double categories of relations
- Having
finite products.

Outline

1. Cartesian bicategories
2. Cartesian equipments
3. Double categories of relations

Target examples

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Bicategory	Rel	Span	Prof
objects (0-cells)	sets $A, B, \dots$	sets $A, B, \dots$	categories $A, B, \dots$
1-cells $A \multimap B$	relations $R \subseteq A \times B$	spans 	profunctors $P: A^{\text{op}} \times B \rightarrow \text{Set}$
2-cells 	inclusion $R \subseteq S (\subseteq A \times B)$	span maps 	transformations 
products (?) & local products	$A \times B$ $R \cap S$	$A \times B$ fiber product 	$A \times B$ $P \times Q$

Cartesian bicategories

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Cartesian bicategories were introduced as a framework to capture the "products" in question and unify **Rel**, **Span**, **Prof**,

HISTORY

- '87 — Carboni, Walters
"Cartesian bicategories I"
- '08 — Carboni, Kelly, Walters, Wood
"Cartesian bicategories II"
- '09? — Todd Trimble's post on nLab.

PROBLEM

The "products" of 0-cells in the previous slide are **NOT** the products in the bicategorical sense.

In **Rel**, $A + B$ is the product of A and B .

$$\frac{C \xrightarrow{R} A + B}{\left(\begin{array}{c} C \xrightarrow{R_1} A \\ C \xrightarrow{R_2} B \end{array} \right)}$$

Cartesian bicategories

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Def ① [CW'87]

A cartesian structure on a locally posetal bicategory $\mathcal{B}$ consists of

- a monoidal product $\otimes$ on $\mathcal{B}$
- a unique cocommutative comonoid structure

$$(\Delta_X : X \longrightarrow X \otimes X, \quad t_X : X \longrightarrow I)$$

for which Δ_X, t_X have right adjoints on every $X \in \mathcal{B}$

s.t. every 1-cell is a lax comonoid homomorphism.

Def A 1-cell in a bicategory is called a map if it allows a right adjoint. $\text{Map}(\mathcal{B}) \subseteq \mathcal{B}$ is the wide locally-full subcategory spanned by maps.

Def (CKWW '08)

A bicategory $\mathcal{B}$ is precartesian if

- (i) the hom-categories $\mathcal{B}(A, B)$ have finite products for all A, B .
- (ii) the bicategory $\text{Map}(\mathcal{B})$ has finite products in the sense of bilimit.

Cartesian bicategories

03 / 21

Def (CW'87)

A cartesian structure on a locally posetal bicategory $\mathcal{B}$ consists of

- a monoidal product $\otimes$ on $\mathcal{B}$

- sufficient structure for $A \otimes B$ to have the universal property w.r.t. "functions".

||

Functional relations

||

Left adjoints in Rel

Def A 1-cell in a bicategory is called a map if it allows a right adjoint. $\text{Map}(\mathcal{B}) \subseteq \mathcal{B}$ is the wide locally-full subcategory spanned by maps.

Def (CKWW '08)

A bicategory $\mathcal{B}$ is precartesian if

- (i) the hom-categories $\mathcal{B}(A, B)$ have finite products for all A, B .
- (ii) the bicategory $\text{Map}(\mathcal{B})$ has finite products in the sense of bilimit.

Cartesian bicategories

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Prop [CKWW'08]

For a precartesian bicategory $\mathcal{B}$,
lax functors $\otimes : \mathcal{B} \times \mathcal{B} \rightarrow \mathcal{B}$ and
 $I : 1 \rightarrow \mathcal{B}$ are induced.

Def ② A precartesian bicategory is
cartesian if $\otimes$ and I above are
pseudo functors.

Thm [CW'87]

The two definitions are equivalent
for locally posetal bicategories.

Todd Trimble presented
an alternative approach on nLab:

Def ③ A **cartesian** structure on $\mathcal{B}$
consists of

- $\otimes : \mathcal{B} \times \mathcal{B} \rightarrow \mathcal{B}$, $I : 1 \rightarrow \mathcal{B}$
- transformations valued on maps
 $\left(\begin{array}{l} \epsilon_A : A \rightarrow I, \delta_A : A \rightarrow A \otimes A, \\ \pi_{A,B} : A \otimes B \rightarrow A, \pi'_{A,B} : A \otimes B \rightarrow B \end{array} \right)$
- invertible modification
with some coherent conditions.

Why is it so hard?

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Problem: the universal properties of the "products" are for maps, not all 1-cells

Remember...

a category $\mathcal{C}$ is **cartesian**
(= has finite products)

$$\text{if } \mathcal{C} \begin{array}{c} \xrightarrow{\Delta} \\ \perp \\ \xleftarrow{\exists \times} \end{array} \mathcal{C} \times \mathcal{C} \quad \& \quad \mathcal{C} \begin{array}{c} \xrightarrow{!} \\ \perp \\ \xleftarrow{\exists 1} \end{array} 1.$$

😊 Easy to check.

😊 Property-like structure.

😊 Only in terms of the 2-cat. **CAT**.

Left adjoints in the bicategory

Def (cartesian objects, CKW'91)

$\mathcal{K}$: 2-category with finite products

$C \in \mathcal{K}$ is a **cartesian object**

$$\text{if } C \begin{array}{c} \xrightarrow{\Delta} \\ \perp \\ \xleftarrow{\exists \times} \end{array} C \times C \quad \& \quad C \begin{array}{c} \xrightarrow{!} \\ \perp \\ \xleftarrow{\exists 1} \end{array} 1$$

It is unknown whether
cartesian bicategories can be formulated
as cartesian objects in any 2-category.

Again, why is it so hard?

It's just finite products for maps!

Maps themselves are not the primitive data of bicategories, making it difficult to formulate universal properties for them.

Then, why don't we separate "maps" from left adjoints
as a new kind of 1-cells,
and define the finite products in terms of these 1-cells?



Double categories !!

Outline

1. Cartesian bicategories
2. Cartesian equipments
3. Double categories of relations

Double categories

Def

A (pseudo) double category consists of

- objects $A, B, \dots$

- vertical arrows $A \downarrow f B, \dots$

- horizontal arrows $A \xrightarrow{p} B, \dots$

- cells $\begin{array}{ccc} A & \xrightarrow{p} & B \\ f \downarrow & \sigma & \downarrow g \\ C & \xrightarrow{q} & D \end{array}, \dots$

with compositions of vertical arrows,

$$\begin{array}{c} A \\ \downarrow f \\ B \\ \downarrow g \\ C \end{array}$$

horizontal arrows,

$$A \xrightarrow{p} K \xrightarrow{m} X$$

and cells

$$\begin{array}{ccccc} A & \xrightarrow{p} & K & \xrightarrow{m} & X \\ f \downarrow & \sigma & \downarrow s & \nu & \downarrow u \\ B & \xrightarrow{q} & L & \xrightarrow{n} & Y \end{array}$$

$$\begin{array}{ccccc} A & \xrightarrow{p} & K & & \\ f \downarrow & \sigma & \downarrow s & & \\ B & \xrightarrow{\quad} & L & & \\ g \downarrow & \tau & \downarrow t & & \\ C & \xrightarrow{r} & M & & \end{array}$$

subject to some coherent conditions.

Double categories

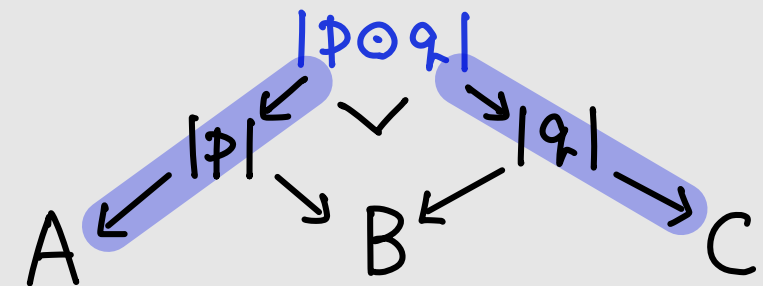
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<u>Example</u>	Rel	Span	Prof
Objects	sets	sets	categories
Vertical arrows	functions	functions	functors
Horizontal arrows	relations	spans	profunctors
Cells	$ \begin{array}{ccc} A & \xrightarrow{p} & B \\ f \downarrow & \tau & \downarrow g \\ C & \xrightarrow{q} & D \end{array} $ A cell exists iff $ \begin{array}{c} p(a, b) \\ \Downarrow \\ q(f(a), g(b)) \end{array} (\forall a, b) $	$ \begin{array}{ccc} A & \xleftarrow{ p } & B \\ f \downarrow & \eta & \downarrow h & \eta & \downarrow g \\ C & \xleftarrow{ q } & D \end{array} $	$ \begin{array}{ccc} A^{\circ p} \times B & \xrightarrow{p} & \text{Set} \\ f^{\circ p} \times g \downarrow & \Downarrow \tau & \\ C^{\circ p} \times D & \xrightarrow{q} & \end{array} $

Remark

The horizontal composition is required to be associative *only up to isomorphism*.

e.g. the composition of spans



Notation

$$\begin{array}{c} A \\ f \swarrow \tau \searrow g \\ B \xrightarrow[p]{} C \end{array} := \begin{array}{c} A \xrightarrow[\tau]{Id_A} A \\ f \downarrow \tau \downarrow g \\ B \xrightarrow[p]{} C \end{array}$$

(e.g. In Rel, the cell implies that $a = a' \implies \mathcal{P}(f(a), g(a')) \quad (\forall a, a')$
 $\implies \mathcal{P}(f(a), g(a)) \quad (\forall a)$)

Similar for $\begin{array}{c} A \xrightarrow[p]{} B \\ f \swarrow \tau \searrow g \\ C \end{array}$.

This convention is justified by AVDCs.
 (See the next talk by Yuto Kawase.)

Equipments

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In Rel, a function f gives rise to a functional relation R_f .

Cells describe this situation as

$$\begin{array}{c} A \\ // \quad \searrow f \\ A \xrightarrow[R_f]{} B \end{array}$$

$$R_f(a, f(a))$$

$$\begin{array}{c} A \xrightarrow[R_f]{} B \\ f \swarrow \tau \searrow \\ B \end{array}$$

$$R_f(a, b) \implies f(a) = b$$

$$+ \begin{array}{c} A \\ // \quad \searrow f \\ A \xrightarrow[R_f]{} B \end{array} = f \left(\begin{array}{c} A \\ \neq \\ B \end{array} \right) f, \quad \begin{array}{c} A \xrightarrow[R_f]{} B \\ // \quad \searrow f \quad // \\ A \xrightarrow[R_f]{} B \end{array} = \begin{array}{c} A \xrightarrow[R_f]{} B \\ // \quad // \quad // \\ A \xrightarrow[R_f]{} B \end{array}$$

Equipments

Def [Shulman '08, Aleiferi '19]

A double category is an **equipment**

if every vertical arrow $\downarrow f$ admits

$A \xrightarrow{f_*} B$ and $B \xrightarrow{f^*} A$ with

$$\begin{array}{c} A \\ \parallel \searrow f \\ A \xrightarrow{f_*} B \end{array} \quad , \quad \begin{array}{c} A \xrightarrow{f_*} B \\ f \searrow \parallel \\ B \end{array} \quad ,$$

s.t.

$$\begin{array}{c} A \\ f \swarrow \parallel \\ B \xrightarrow{f^*} A \end{array} \quad , \quad \begin{array}{c} B \xrightarrow{f^*} A \\ \parallel \swarrow f \\ B \end{array}$$

$$\begin{array}{c} A \\ \parallel \searrow f \\ A \xrightarrow{f_*} B \end{array} = f \left(\begin{array}{c} A \\ \parallel \searrow f \\ B \end{array} \right) f \quad \&$$

$$\begin{array}{c} A \xrightarrow{f_*} B \\ \parallel \searrow f \\ A \xrightarrow{f_*} B \end{array} = \begin{array}{c} A \xrightarrow{f_*} B \\ \parallel \quad \parallel \quad \parallel \\ A \xrightarrow{f_*} B \end{array}$$

and their
duals hold.

Example

In Prof, for $F: \mathcal{C} \rightarrow \mathcal{D}$

$$F^*(\bullet, \circ) := \mathcal{D}(F\bullet, \circ): \mathcal{C} \rightarrow \mathcal{D}$$

$$F_*(\circ, \bullet) := \mathcal{D}(\circ, F\bullet): \mathcal{D} \rightarrow \mathcal{C}$$

Restrictions in equipments

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Example

For $f \downarrow A \xrightarrow{R} B$ and $g \downarrow B' \xrightarrow{p} B'$ in $\mathbf{Rel}$,

there is the largest relation $A' \xrightarrow{S} B'$

$$\text{s.t. } \begin{array}{ccc} A' & \xrightarrow{S} & B' \\ f \downarrow & \cap & \downarrow g \\ A & \xrightarrow{R} & B \end{array}$$

Namely,

$$R[f;g](a,b) \stackrel{\text{def}}{\iff} R(f(a), g(b)).$$

This is achieved with the composite

$$A' \xrightarrow{f_*} A \xrightarrow{R} B \xrightarrow{g^*} B'.$$

Def

A restriction of $f \downarrow A \xrightarrow{p} B$ and $g \downarrow B' \xrightarrow{q} C$ is

$$A' \xrightarrow{p[f;g]} B' \text{ together with } \begin{array}{ccc} A' & \xrightarrow{p[f;g]} & B' \\ f \downarrow & \text{cart} & \downarrow g \\ A & \xrightarrow{p} & B \end{array}$$

that shows the following universal property:

$$\begin{array}{ccc} X & \xrightarrow{q} & Y \\ x \downarrow & & \downarrow y \\ A' & \xrightarrow{\exists} & B' \\ f \downarrow & & \downarrow g \\ A & \xrightarrow{p} & B \end{array} = \exists! \begin{array}{ccc} X & \xrightarrow{q} & Y \\ x \downarrow & & \downarrow y \\ A' & \xrightarrow{\exists} & B' \\ f \downarrow & \text{cart} & \downarrow g \\ A & \xrightarrow{p} & B \end{array}.$$

Restrictions in equipments

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Prop

A double category is an equipment
iff every $f \downarrow A \xrightarrow{p} B \downarrow g$ has a restriction.

proof

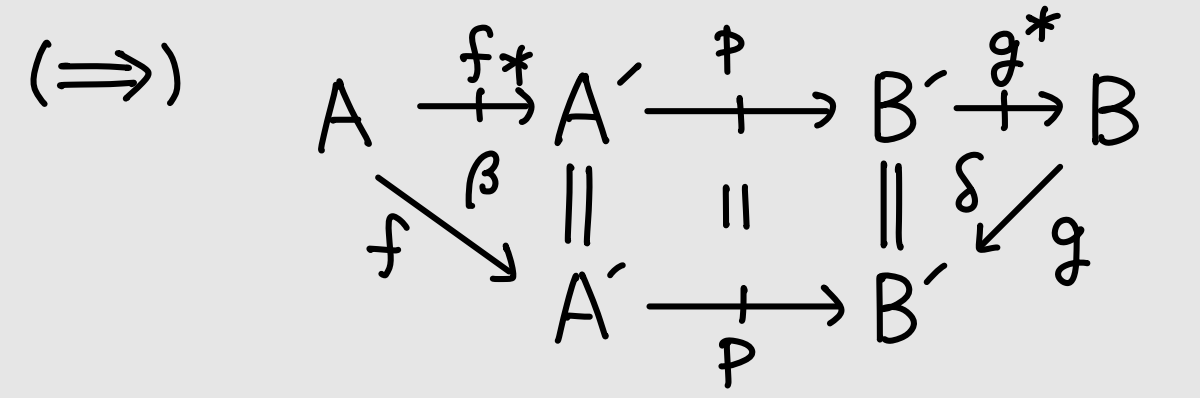
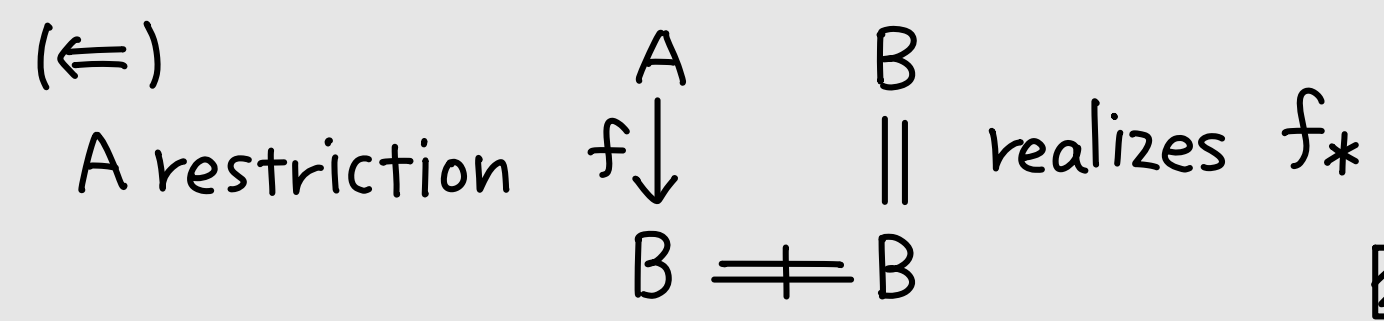
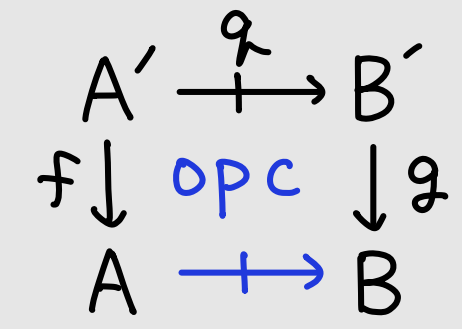


exhibit $f_* p g^*$ as a restriction.



Remark

This is also equivalent to
the vertical dual condition.



We now turn to define
"equipments with finite products."

Cartesian equipments

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Def (cartesian objects, CKW'91)

$\mathcal{K}$: 2-category with finite products

$C \in \mathcal{K}$ is a cartesian object

if

$$C \begin{array}{c} \xrightarrow{\Delta} \\ \perp \\ \xleftarrow{\epsilon} \end{array} C \times C \quad \& \quad C \begin{array}{c} \xrightarrow{!} \\ \perp \\ \xleftarrow{\epsilon} \end{array} 1$$

Let $\mathbf{Eqp}$ be the 2-cat. of equipments, double functors, and vertical transformation.

With this, we can express the universal property with respect to vertical arrows.

A vertical transformation.

$\alpha : F \Rightarrow G : \mathbb{D} \rightarrow \mathbb{E}$ consists of

$$\left(\begin{array}{c} FA \\ \alpha_A \downarrow \\ GA \end{array} \right)_{A \in \mathbb{D}} \quad \& \quad \left(\begin{array}{ccc} FA & \xrightarrow{Fp} & FB \\ \alpha_A \downarrow & \alpha_p & \downarrow \alpha_B \\ GA & \xrightarrow{Gp} & GB \end{array} \right)_{A \xrightarrow{p} B}$$

with the naturality condition.

Cartesian equipments

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Def (cartesian objects, CKW '91)

$\mathcal{K}$: 2-category with finite products

$C \in \mathcal{K}$ is a cartesian object

if

$$C \begin{array}{c} \xrightarrow{\Delta} \\ \perp \\ \xleftarrow{\epsilon \times} \end{array} C \times C \quad \& \quad C \begin{array}{c} \xrightarrow{!} \\ \perp \\ \xleftarrow{\epsilon} \end{array} 1$$

Let $\mathbf{Eqp}$ be the 2-cat. of equipments, double functors, and vertical transformation.

With this, we can express the universal property with respect to vertical arrows.

Def [Aleiferi '19]

A cartesian equipment is a cartesian object in $\mathbf{Eqp}$.

Prop (Unraveling the data) [Aleiferi '19]

An equipment $\mathbb{D}$ is cartesian iff

- (i) its vertical category is cartesian,
- (ii) the horizontal hom-categories $\mathbb{D}(A, B)$ are cartesian for $A, B \in \mathbb{D}$,
- (iii) The lax functors induced by (i), (ii)
 $\mathbb{D} \times \mathbb{D} \xrightarrow{\times} \mathbb{D}, \quad 1 \xrightarrow{!} \mathbb{D}$
are double (= pseudo) functors.

More examples of cartesian equipments

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- $\mathbf{Rel}(\mathcal{E})$
 $\mathcal{E}$: a regular category
- $\mathbf{Span}(\mathcal{E})$
 $\mathcal{E}$: a category w/ finite limits
- $\mathbf{Rel}_{(E,M)}(\mathcal{E})$ [HN.'25]
 (E,M) : an SOFS on $\mathcal{E}$

$$\frac{A \xrightarrow{R} B}{(R \rightarrow A \times B) \in M}$$

- $\mathbf{Prof}(S)$ (internal profunctors)
 S : a category w/ finite limits and stable coequalizer.
- $S\text{-}\mathbf{Prof}$ (enriched profunctors)
 S : a cartesian category w/ coequalizers preserved by $(A \times -)$'s.
- $\mathbf{Lex}$: obj.: categories w/ finite limits

$$\begin{array}{ccc} A \xrightarrow{k} B & & A \xrightarrow{k} B \\ F \downarrow \tau \downarrow G & \parallel & F \left(\begin{array}{c} \uparrow \\ \downarrow \end{array} \right) \tau \downarrow G \left(\begin{array}{c} \uparrow \\ \downarrow \end{array} \right) \\ C \xrightarrow{l} D & & C \xrightarrow{l} D \end{array} \quad \text{fin. lim. preserving functors}$$

Merits in cartesian equipments

- Easy to check the condition

Example Rel is cartesian.

proof

$$\text{Rel} \times \text{Rel} \xrightarrow[\times]{\Delta} \text{Rel}$$

$$\begin{array}{ccc} A \xrightarrow{R} B & A' \xrightarrow{R'} B' & A \times A' \xrightarrow{R \times R'} B \times B' \\ (f \downarrow \quad \cap \downarrow g, f' \downarrow \quad \cap \downarrow g') \mapsto & & f \times f' \downarrow \quad \cap \downarrow g \times g' \\ C \xrightarrow{S} D & C' \xrightarrow{S'} D' & C \times C' \xrightarrow{S \times S'} D \times D' \end{array}$$

$$\begin{array}{ccc} X \xrightarrow{P} Y & X \xrightarrow{P} Y & X \xrightarrow{P} Y \\ (s \downarrow \quad \cap \downarrow t, s' \downarrow \quad \cap \downarrow t') \parallel \langle s, s' \rangle \downarrow \quad \cap \downarrow \langle t, t' \rangle & & A \times A' \xrightarrow{R \times R'} B \times B' \\ A \xrightarrow{R} B & A' \xrightarrow{R'} B' & \end{array}$$

- Well-behaved with some constructions

The general theory of cartesian objects helps us construct a new cartesian equipments.

Example

$\text{Prof} = \text{Mod}(\text{Span})$ is a cartesian equipment.

proof

$\text{Eqp}^* \xrightarrow{\text{Mod}} \text{Eqp}$ preserves $\times$ and 1

$\Rightarrow$ lifts to $\text{Eqp}_{\text{cart}}^* \xrightarrow{\text{Mod}} \text{Eqp}_{\text{cart}}$.

(We need to assume local coequalizers for equipments that we are applying Mod to. $*$ is for this condition.)

- Flexible

We no longer need to take maps as vertical arrows.

Example Maps in Prof need not be of form F_* .

They are "functors up to Cauchy completion".

Cartesian bicategories VS cartesian equipments

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Q. Is there a cartesian equipment
whose horizontal part is B ?
(B : cartesian bicategory)

A. Almost.

Verity's PhD thesis presents

a double bicategorical $\text{Map}(B)$
cartesian equipment
whose horizontal part is B .

(This is inevitable due to the coherence
problem without further assumption
[N. '25, Lemma 2.5.15].)

Q. Is the horizontal part of $\mathbb{D}$
a cartesian bicategory?
($\mathbb{D}$: cartesian equipment)

A. Yes. [Patterson '24, N. '25]

Diagrammatic proof
Abstract proof based on Trimble's definition.

Remark

This theorem states that $A \times B$
in a cart. eqp. has the universal property
not only among vertical arrows
but also among maps of horizontal arrows.

Outline

1. Cartesian bicategories
2. Cartesian equipments
3. Double categories of relations

Double categories of relations

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Double categories that serve as semantic environments
for predicate logic by modeling binary relations

○	×
$\text{Rel}(\mathcal{E})$	$\text{Prof}(S)$
$\text{Span}(\mathcal{E})$	$S\text{-Prof}$
$\text{Rel}_{(E,M)}(\mathcal{E})$	$\mathbb{L}ex$

What set these apart?

“Symmetry/involution” ?

“Horizontal arrows $A \xrightarrow{p} B$ are ?
only dependent on $A \times B$ ”

Definition 2.1. (i) An object X in a Cartesian bicategory is *discrete* when the multiplication Δ_X^* and the comultiplication Δ_X satisfy

$$(D) \quad \Delta \cdot \Delta^* = (\Delta^* \otimes 1) \cdot (1 \otimes \Delta).$$

(We are forgetting the associativity in the middle.)

(ii) A Cartesian bicategory is called a '*bicategory of relations*' if every object is discrete. [Carboni, Walters. '87]

This condition was later rephrased as *the Frobenius law*, because this is equivalent to the condition for Frobenius algebra [WW '08].



The Frobenius law for cartesian bicategories state that

$$\exists a \quad \left(\begin{array}{ccc} x & \rightsquigarrow & z \\ y & \rightsquigarrow & w \end{array} \right) a \rightsquigarrow \left(\begin{array}{ccc} x & \rightsquigarrow & z \\ y & \rightsquigarrow & w \end{array} \right) \iff \begin{array}{ccc} x & \rightsquigarrow & z \\ y & \rightsquigarrow & w \end{array}$$

(When $A \xrightarrow{\text{id}} A$ is understood as $\{(a, a') \mid a \rightsquigarrow a'\}$)

The Frobenius law

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The Frobenius law for bicategories / equipments appears in the literature to characterize a certain classes of those bi/double categories.

[CW'88, WW'08, LWW'10, Aleiferi'19, Lambert'23, HN'25]

What does the Frobenius law actually mean?

Prop [WW'08, HN'25, N.25]

A Frobenius cartesian equipment

always comes with a (horizontal) self-dual compact closed structure w.r.t. $\times$

$$\frac{A \times B \multimap C}{A \multimap B \times C}$$

on the horizontal bicategory
in the sense of [Stay'13].

The Frobenius law and fibrations

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Thm [N. 25]

A Frobenius cart. eqp. $\mathbb{D}$ induces an elementary existential fibration $\mathcal{U}_{\mathbb{D}} : \mathcal{U}_{\mathbb{D}} \downarrow \mathbb{D}_0$.

the base category = the vertical category $\mathbb{D}_0$

the fiber over A = the category of horizontal arrows of the form $A \xrightarrow{P} 1$

and cells $\begin{array}{ccc} A & \xrightarrow{P} & 1 \\ \parallel & \tau & \parallel \\ A & \xrightarrow{Q} & 1 \end{array}$ between them.

cartesian lifting along $\begin{array}{c} A \\ \downarrow f \\ B \end{array}$ = restriction $\begin{array}{ccc} A & \xrightarrow{f^*P} & 1 \\ f \downarrow \text{cart} & & \parallel \\ B & \xrightarrow{P} & 1 \end{array}$

Remark What is not obvious is the Frobenius reciprocity for $\mathcal{U}_{\mathbb{D}}$.

The Frobenius law and fibrations

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Thm [N. 25]

A Frobenius cart. eqp. $\mathbb{D}$ induces an elementary existential fibration

$$\begin{array}{c} \mathcal{U}_{\mathbb{D}} \\ \downarrow \mathcal{U}_{\mathbb{D}} \\ \mathbb{D}_0 \end{array}$$

Thm [N. 25] We have an equivalence $\mathbf{Eqp}_{\text{cart, Frob}} \xrightarrow[\simeq]{\mathcal{U}} \mathbf{EEF}$.

Remark Its inverse is $\mathbf{EEF} \xrightarrow{\text{Bil}} \mathbf{Eqp}_{\text{cart}}$ in [N. 25].

$$\begin{array}{ccc} \begin{array}{ccc} A & \xrightarrow{R} & B \\ f \downarrow & \tau & \downarrow g \\ C & \xrightarrow{S} & D \end{array} & \text{in } \mathbf{Bil}(\mathcal{P}) & \parallel & \begin{array}{ccc} R & \xrightarrow{\tau} & S \\ \downarrow & & \downarrow \\ A \times B & \xrightarrow{f \times g} & C \times D \end{array} & \text{in } \begin{array}{c} \mathcal{E} \\ \downarrow \mathcal{P} \\ B \end{array} \end{array}$$

See SideA of this talk.

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Thank you!

my homepage :

hayatonasu.github.io

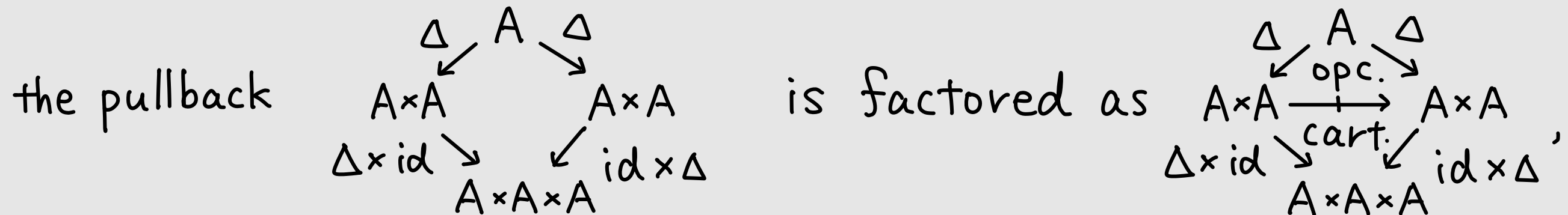
my thesis :

[arXiv:2501.17869](https://arxiv.org/abs/2501.17869)

Appendix: the Frobenius law in equipments

A 1 /

In terms of cartesian equipments, the Frobenius law can be stated as :



(the Beck-Chevalley pullback [HN. 25].)

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Cartesian equipments

the Frobenius law